Question #1 of 120

Question ID: 1214547

Questions 1 through 18 relate to Ethical and Professional Standards. (27 minutes)

Ray Brown, CFA, gives prospects his firm's marketing materials, not prepared by him, that indicate he has a graduate degree from State University, when in fact he did graduate work there but did not receive a degree. Brown informed the marketing department of this error when he first saw it. Brown has:

A) violated the Standards by misrepresenting his qualifications.

B) not violated the Standards because he has informed his firm of the mistake.

C) not violated the Standards because he did not prepare the marketing materials or misrepresent his credentials to his firm.

Explanation

Brown has violated Standard I(C) Misrepresentation by giving prospects firm marketing materials that he knows are incorrect.

For Further Reference:

(Study Session 1, Module 3.2, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #2 of 120

Question ID: 1214562

John Larsen, CFA, is creating his investment firm's initial GIPS-compliant performance results. He would like to supplement the historical performance numbers with older, non-GIPS-compliant data. According to GIPS, is this allowed?

A) GIPS results cannot include presentation of any noncompliant performance data.

B) After the initial, GIPS-compliant performance results are presented, a firm may go back further and present non-compliant performance data, but no non-compliant results can be included for time periods after January 1, 2000.

C) As long as five years of GIPS-compliant performance results are presented, the firm can go back further and present non-compliant performance data.

Explanation

GIPS-compliant results can be presented with non-compliant historical performance added for earlier periods, but no non-compliant results can be presented for any time period after January 1, 2000.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.b)

CFA® Program Curriculum, Volume 1, page 236

Question #3 of 120

Question ID: 1214556

Jenny Pickler, a Level II CFA Candidate, writes an economic forecast containing several interest rate projections. Her firm's investment committee reviews Pickler's report and changes several of the interest rates Pickler had forecast. To comply with CFA Institute Standards, Pickler:

A) **does not need to take any further action.**
B) should ask that her name be removed from the report.
C) is required to independently review the data supporting the investment committee's changes.

Explanation

According to Standard V(A) Diligence and Reasonable Basis, group consensus is not required in the course of preparation of analytical reports. Pickler would only need to have her name removed from the report if she had reason to believe the investment committee did not have a reasonable and adequate basis for their changes.

For Further Reference:

(Study Session 1, Module 3.7, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #4 of 120

Question ID: 1214546

Which of the following statements is *most accurate* about the Standard concerning knowledge of the law? Members and candidates are responsible for violations:

A) that are proven by a regulatory authority.
B) of which they are aware or should be aware.
C) **in which they knowingly participate or assist.**

Explanation

Standard I(A) Knowledge of the Law holds members and candidates responsible for violations in which they knowingly participate or assist. The Standard is not limited to violations that are proven by a regulatory authority. Although the Standard strongly encourages members and candidates to report potential violations of which they are aware, the Standard does not require them to do so.

For Further Reference:

(Study Session 1, Module 3.1, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #5 of 120

Question ID: 1214549

Andrew Pollard, CFA, overhears two executives from a multinational oil company discussing an unexpectedly large earnings increase the company is preparing to announce the following morning. When Pollard gets home that evening, he places an order to buy shares in this oil company. Which of the following *best* describes this situation?

A) **Pollard violated the Standards by acting on material nonpublic information.**

B) There is no violation of CFA Institute Standards, since this was simply an overheard conversation.

C) Pollard violated CFA Institute Standards by not contacting counsel for advice before placing the trade.

Explanation

Pollard has enough information to determine that the overheard information is indeed material nonpublic information. No matter how this information was obtained, even through an overheard conversation, Pollard may not act or cause others to act on it. Even if he had contacted internal counsel before placing the trade, Pollard would have violated Standard II(A) Material Nonpublic Information.

For Further Reference:

(Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #6 of 120

Question ID: 1214548

Recommended procedures for compliance with the Standard concerning misconduct suggest that firms in the investment industry should:

A) periodically test their employees' knowledge of applicable laws, regulations, and the firm's code of ethics.

B) periodically inform employees of violations that have occurred and the disciplinary actions that the firm took against the employees involved.

C) **check references of potential employees to verify that they are of good character and eligible for employment in the investment industry.**

Explanation

Checking the references given by potential employees is one of the recommended procedures for compliance with Standard I(D) Misconduct. Other recommended procedures are that the firm adopt a code of ethics and inform employees of potential violations and their consequences for disciplinary action. Neither testing employees' knowledge of laws and regulations nor informing them of actual violations by other employees is specified as a recommended procedure.

For Further Reference:

(Study Session 1, Module 3.2, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #7 of 120

Question ID: 1214557

Which of the following statements is *most accurate*? An analyst who changes employers and wants to maintain coverage of a stock:

A) may copy supporting records from the prior firm and use them at the new firm.

B) **must re-create the supporting records at the new firm with information from public sources or from the covered firm.**

C) may maintain his recommendations at the new firm without re-creating the supporting documentation if those recommendations had a reasonable basis.

Explanation

Supporting records for analyst recommendations are the property of the analyst's firm. A member or candidate who changes firms may not take these records without permission. To continue coverage of the same securities at a new firm, a member or candidate must re-create the supporting records based on information from the covered firm or from publicly available sources.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #8 of 120

Question ID: 1214559

After taking the Level I exam, Willie Winchester posts a comment on a social media website wondering why the exam had no questions about interest rate risk and currency risk. Winchester has:

A) violated the Code and Standards by discussing the CFA Program on social media.

B) **violated the Code and Standards by disclosing confidential information about the exam.**

C) not violated the Code and Standards because he did not disclose specific exam questions.

Explanation

Winchester violated Standard VII(A) Conduct as Participants in CFA Institute Programs by revealing topics that were not tested on the exam. Candidates are not restricted from discussing the CFA Program in general.

For Further Reference:

(Study Session 1, Module 3.9, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #9 of 120

Question ID: 1214552

A member provides a client with an investment performance presentation that does not include detailed information, but reflects the member's reasonable efforts to present results that are fair, accurate, and complete. Has the member complied with the Standard related to performance presentation?

A) **Yes, the member has met the requirements of the Standard.**

B) No, because the performance presentation must comply with Global Investment Performance Standards.

C) No, because “reasonable efforts” do not ensure that the presentation is fair, accurate, and complete.

Explanation

Brief presentations are acceptable if they include a statement that detailed information is available upon request. Standard III(D) Performance Presentation requires members and candidates to make reasonable efforts to ensure fair, accurate and complete presentation of results. While compliance with GIPS is recommended to meet Standard III(D) obligations, use of GIPS is not required.

For Further Reference:

(Study Session 1, Module 3.5, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #10 of 120

Question ID: 1214554

With regard to independent practice by Members and Candidates who are employed, the Code and Standards specify that:

- **A)** undertaking independent practice includes preparations to begin such practice.
- **B)** written consent must be obtained from both the employer and clients who may be affected.
- **C)** **members and candidates contemplating independent competitive business must notify their current employer of the types of services to be rendered, duration, and compensation.**

Explanation

Standard IV(A) Loyalty requires that members and candidates notify their employer all details of the independent practice and receive the employer's consent before engaging in the competitive activity, but does not require any statement from firm clients. If the independent practice is likely to affect clients negatively, the employer can refuse permission. Making preparations to begin a competitive practice is allowed, as long as it does not interfere with current employment duties.

For Further Reference:

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #11 of 120

Question ID: 1214560

Ken Koski, CFA, issues a press release that includes the following statement:

> We are proud to announce that two of our managers have earned the right to use the CFA designation. In addition, four of our junior analysts have become Level III CFA candidates. These individuals have proven their dedication to the investment community and shown commitment to the highest ethical standards.

With regard to the statements in the press release:

- **A)** **all these statements are in compliance with CFA Institute Standards.**
- **B)** Koski has violated the Code and Standards by improperly referencing the managers' right to use the CFA designation.
- **C)** Koski has violated the Code and Standards by implying superior performance results.

Explanation

All of the statements are acceptable according to Standard VII(B), Reference to CFA Institute, the CFA designation, and the CFA Program. Koski is allowed to make a statement of fact such as the managers' right to use the CFA designation. Koski may reference the participation of his employees in the CFA program if the employees are currently registered to take one of the exams. The statements regarding dedication to the investment community and commitment to the highest ethical standards are proper references regarding the CFA program.

For Further Reference:

(Study Session 1, Module 3.9, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #12 of 120

Question ID: 1214550

Gary Hoskins, CFA, runs a macro strategy hedge fund. The hedge fund has a short position in July osmium futures. Hoskins attempts to gain control of the available supply of osmium that can be delivered in July, which will cause holders of long July futures to make offsetting trades instead of settling by delivery. Has Hoskins violated the Standard concerning market manipulation?

A) **Yes.**
B) No, because Hoskins is executing an arbitrage trade.
C) No, because Hoskins is not manipulating the price of osmium.

Explanation

Hoskins is attempting to manipulate the price of a derivative by securing a controlling position in the underlying asset, which is a violation of Standard II(B) Market Manipulation.

For Further Reference:

(Study Session 1, Module 3.3, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #13 of 120

Question ID: 1214563

Ed Socho states that in a GIPS-compliant presentation, (1) total firm assets must include all accounts, including non-fee-paying accounts and accounts where the client makes the investment decisions, and (2) the firm must include the performance results of third-party advisors selected by the firm in its composite performance. Are Socho's statements accurate?

A) **Both of these statements are accurate.**
B) Neither of these statements is accurate.
C) Only one of these statements is accurate.

Explanation

Both statements are correct. Total firm assets must include fee-paying and non-fee-paying accounts. If a sub-advisor who manages firm assets is selected by the firm, the performance of assets under the sub-advisor's control must be included in the performance of the firm's composite for those assets.

For Further Reference:

(Study Session 1, Module 5.1, LOS 5.a)

CFA® Program Curriculum, Volume 1, page 235

Question #14 of 120

Question ID: 1214558

Isaac Jones, CFA, is a portfolio manager for a major brokerage firm. Jones wishes to buy Maxima common stock for some of his clients' accounts. Jones also wishes to purchase Maxima for his personal account. In accordance with CFA Institute Standards, Jones may purchase Maxima for his personal account:

A) **only after completing the transactions for his clients.**

B) along with the purchases for his clients, as long as this is disclosed in advance to his clients and employer.

C) at any time, as long as the execution price is not more favorable than the execution price received by the clients.

Explanation

In accordance with Standard VI(B) Priority of Transactions, employer and client transactions must take priority over any personal transactions, meaning any transactions in which the member or candidate is the beneficial owner. Disclosure is not enough to comply with this Standard and the execution price is not relevant.

For Further Reference:

(Study Session 1, Module 3.8, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #15 of 120

Question ID: 1214561

Which of the following statements *best* describes how GIPS requires portfolios to be grouped into composites?

A) Composites can include model results, if this is clearly specified.

B) All discretionary portfolios must be included in at least one composite if they are still managed by the firm.

C) **Each composite must include all discretionary portfolios that the firm has managed according to that particular composite strategy or style, including closed accounts.**

Explanation

All discretionary portfolios, whether closed or not, must be included in composite results for the period they were managed. Model results may not be included in composite results.

For Further Reference:

(Study Session 1, Module 4.1, LOS 4.b)

CFA® Program Curriculum, Volume 1, page 229

Question #16 of 120

Question ID: 1214555

With respect to the responsibilities of supervisors, the Code and Standards state that those with supervisory responsibility:

A) may not delegate supervisory responsibility.

B) are in violation if an employee under their supervision commits securities fraud.

C) must institute procedures to prevent and detect violations of rules and regulations by those subject to their supervision.

Explanation

Standard IV(C) Responsibilities of Supervisors requires members and candidates with supervisory responsibility to make reasonable efforts to detect and prevent violations of rules and regulations (as well as of the Code and Standards) by those under their supervision. The fact that violations occur is not necessarily evidence that reasonable efforts were not made. In large organizations, delegating supervisory responsibility may be necessary, but this does not relieve the person with overall authority of supervisory responsibility.

For Further Reference:

(Study Session 1, Module 3.6, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #17 of 120

Question ID: 1214553

The Standard concerning suitability recommends that the objectives and constraints of an investment policy statement should be reviewed at least:

A) annually.
B) twice each year.
C) every two years.

Explanation

Recommendations for compliance with Standard III(C) Suitability state that an investment policy statement should be reviewed at least annually. The recommendations also note that changes in market conditions or client circumstances may make more frequent updates necessary.

For Further Reference:

(Study Session 1, Module 3.5, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #18 of 120

Question ID: 1214551

A member or candidate who changes his recommendation on a stock can comply with the Standards by communicating this change to clients according to:

A) size of the client.
B) known interest of the client in the stock.
C) number of shares of the stock owned by the client.

Explanation

Although simultaneous distribution of information is preferred, distributing recommendations, or changes of recommendations, first to those clients who have previously expressed interest in these types of securities is acceptable. Giving preferred treatment to larger accounts would violate Standard III(B) Fair Dealing.

For Further Reference:

(Study Session 1, Module 3.4, LOS 3.a, 3.b, 3.c)

CFA® Program Curriculum, Volume 1, page 55

Question #19 of 120

Question ID: 1214567

Questions 19 through 30 relate to Quantitative Methods. (18 minutes)

A recent study indicates that the probability that a company's earnings will exceed consensus expectations equals 50%. From this analysis, the odds that the company's earnings exceed expectations are:

A) 1 to 2.

B) 2 to 1.

C) 1 to 1.

Explanation

Odds for an event equals the ratio of the probability of success to the probability of failure. If the probability of success is 50%, then there are equal probabilities of success and failure, and the odds for success are 1 to 1.

For Further Reference:

(Study Session 2, Module 8.1, LOS 8.c)

CFA® Program Curriculum, Volume 1, page 460

Question #20 of 120

Question ID: 1214564

Sydney Burns, CFA, is considering the purchase of a bond issued by SubPrime Providers. The bond is highly liquid and has a maturity equal to that of a long-term Treasury bond. The SubPrime Providers bond carries a default risk premium of 5%. Burns notices that the difference in interest rates offered on long-term Treasury bonds and short-term Treasury bills currently equals 4%. The real risk-free rate equals 1% and the expected inflation rate equals 2%. Burns should expect the interest rate on the SubPrime Providers bond to:

A) be greater than or equal to 4%, and less than or equal to 8%.

B) be greater than or equal to 5%, and less than or equal to 9%.

C) be greater than or equal to 7%, and less than or equal to 12%.

Explanation

The interest rate equals the sum of the real rate, the expected inflation rate, the total risk premium (which equals the sum of the maturity risk premium, the liquidity risk premium, and the default risk premium). The real rate equals 1%, the expected inflation rate equals 2%, the maturity risk premium equals 4%. Treasury bonds have no liquidity or default risk, so the interest rate on a long-term Treasury bond would be expected to equal 7%. Since the SubPrime Providers (long-term) bond incurs default risk, its interest rate must exceed that of the long-term Treasury bond (i.e., 7%). The SubPrime Providers bond is highly liquid, so it has no liquidity premium. The default risk premium for SubPrime Providers bond equals 5%. Taken together, this implies that the SubPrime Provider bond interest rate should exceed the Treasury bond interest by 5 percentage points (12%).

For Further Reference:

(Study Session 2, Module 6.1, LOS 6.b)

CFA® Program Curriculum, Volume 1, page 315

Question #21 of 120

Question ID: 1214565

An analyst has calculated the arithmetic, harmonic, and geometric mean using the last 10 years of returns on a stock. Which of these means should the analyst *most appropriately* use to forecast next year's return on the stock?

A) Harmonic mean.
B) Geometric mean.
C) **Arithmetic mean.**

Explanation

The arithmetic mean is statistically the best estimate (expected value) of the next year's return. The harmonic mean is not typically used to compute the historical performance or forecast the expected performance of an investment; rather it is used to compute the average cost of shares purchased over time. The geometric mean is used to calculate average annual compound returns. It is the best estimate of future multi-year annual compound returns, but the arithmetic mean is the best estimate of a single year's return.

For Further Reference:

(Study Session 2, Modules 7.2, 7.3, LOS 7.e, 7.m)

CFA® Program Curriculum, Volume 1, page 386

CFA® Program Curriculum, Volume 1, page 434

Question #22 of 120

Question ID: 1214566

The histogram of returns data for the Accel Equity Fund has a long left tail and is more peaked than a normal distribution. Based on the histogram, the distribution of returns for Accel has:

A) positive skewness.
B) **negative skewness.**
C) negative excess kurtosis.

Explanation

The histogram contains a long left tail, which indicates significant negative skew for the distribution. If the histogram contained a long right tail, the distribution would have exhibited positive skew. A distribution with negative excess kurtosis (i.e., a platykurtic distribution) is less peaked and has thinner tails compared to a normal distribution. A distribution that is more peaked and has thicker tails compared to a normal distribution has positive excess kurtosis (i.e., a leptokurtic distribution).

For Further Reference:

(Study Session 2, Modules 7.1, 7.3, LOS 7.d, 7.j)

CFA® Program Curriculum, Volume 1, page 381

CFA® Program Curriculum, Volume 1, page 424

Question #23 of 120

Question ID: 1214572

A study finds that stocks with low price-to-book-value ratios, using end-of-year stock prices and book values per share, have positive abnormal returns in January on average. This study *most likely* suffers from:

A) **look-ahead bias.**
B) time-period bias.
C) sample selection bias.

Explanation

Look-ahead bias occurs when a study examines an effect based on information that was not yet available at the time being tested. In this case, year-end book values per share are not known until well into the first quarter of the following year.

Time-period bias is present when a study covers either too short a period (the proposed relationship may only hold during that time frame) or too long a period (the proposed relationship may have changed during that span). Sample selection bias refers to taking a sample that is not representative of the population being studied.

For Further Reference:

(Study Session 3, Module 10.2, LOS 10.k)

CFA® Program Curriculum, Volume 1, page 589

Question #24 of 120

Question ID: 1214575

Jane Wilcott, CFA, is researching whether value stocks can be expected to outperform growth stocks in any given month. Examining 10 years of monthly returns on a value stock portfolio and a growth stock portfolio, Wilcott records a positive sign for any month the return on the value portfolio exceeded that of the growth portfolio, and a negative sign for any month the return on the value portfolio was less than that of the growth portfolio. Wilcott tests the null hypothesis that the number of positive months is less than or equal to the number of negative months. Wilcott's research design is an example of a:

A) paired comparisons test.
B) conditional test.
C) **nonparametric test.**

Explanation

Nonparametric tests often transform the original data into ranks or signs. The null hypotheses are then stated in terms of ranks or signs.

For Further Reference:

(Study Session 3, Module 11.3, LOS 11.l)

CFA® Program Curriculum, Volume 1, page 643

Question #25 of 120

Question ID: 1214574

Pat Harris, CFA, examines earnings data for 3 energy companies:

Company	Average per-share quarterly earnings	*p*-value

Axxon Industries	$2.00	0.25
Babson Drilling	$0.50	0.04
Cerex Energy	$3.00	0.01

Harris is asked to test the hypothesis for each company that mean earnings equal zero. Using a 5% level of significance, Harris should conclude that the null hypothesis should be rejected for:

A) Babson only.
B) Axxon and Cerex only.
C) Babson and Cerex only.

Explanation

The *p*-value equals the probability that the null hypothesis is true. For instance, the data indicate that there is a 25% probability that the Axxon Industries (population) earnings equal zero, a 4% probability that the Babson Drilling earnings equal zero, and a 1% probability that the Cerex Energy earnings equal zero. The decision rule is to reject the null hypothesis whenever the *p*-value is less than the significance level. Therefore, the null hypothesis should be rejected for Babson Drilling and Cerex Energy, but not for Axxon Industries.

For Further Reference:

(Study Session 3, Module 11.2, LOS 11.f)

CFA® Program Curriculum, Volume 1, page 619

Question #26 of 120

Question ID: 1214569

An analyst is creating a binomial tree for a stock price based on the following assumptions: the probability of an up move is 62%, the up factor is 1.15, and the down factor is 1/1.15. The probability that the stock price will be unchanged after two periods is *closest* to:

A) 24%.
B) 42%.
C) 47%.

Explanation

The stock price will be unchanged after two periods for either an up-move followed by a down-move or a down-move followed by an up-move. The probability of each of these outcomes is 0.62 (1 – 0.62) = 0.2356 and the probability that one of these outcomes will occur is 2 × 0.2356 = 0.4712.

For Further Reference:

(Study Session 3, Module 9.1, LOS 9.g)

CFA® Program Curriculum, Volume 1, page 526

Question #27 of 120

Question ID: 1214571

The continuously compounded annual rate of return that would increase the value of an investment by 20% in three years is *closest* to:

A) 5.7%.
B) 6.1%.

C) 6.7%.

Explanation

The continuously compounded annual rate of return is ln(1.20) / 3 = 0.0608 or 6.08%.

For Further Reference:

(Study Session 3, Module 9.3, LOS 9.o)

CFA® Program Curriculum, Volume 1, page 541

Question #28 of 120

Question ID: 1214568

On the loading dock of an automobile shipping company, there are 44 blue cars, of which 19 have driver-assist technology; 22 red cars with driver-assist technology; and 34 red cars without driver assist technology. The probability that a car without driver-assist technology is blue is:

A) **42.4%.**
B) 56.8%.
C) 59.0%.

Explanation

44 – 19 = 25 blue cars do not have driver-assist technology, and 34 red cars do not have driver-assist technology, for a total of 25 + 34 = 59 cars that do not have driver-assist technology. The probability that a car without driver-assist technology is blue is 25 / 59 = 42.4%.

For Further Reference:

(Study Session 2, Module 8.1, LOS 8.f)

CFA® Program Curriculum, Volume 1, page 463

Question #29 of 120

Question ID: 1214573

To select the correct Student's *t*-distribution for a hypothesis test, a researcher must identify its:

A) mean and variance only.
B) **degrees of freedom only.**
C) mean, variance, and degrees of freedom.

Explanation

Each Student's *t*-distribution is defined by a single parameter, its degrees of freedom.

For Further Reference:

(Study Session 3, Module 10.2, LOS 10.i)

CFA® Program Curriculum, Volume 1, page 584

Question #30 of 120

Question ID: 1214570

A continuous probability distribution is bounded by –5 and +5. Its cumulative distribution function is:

A) unbounded.

B) **bounded by 0 and 1.**

C) bounded by –5 and +5.

Explanation

A cumulative distribution function gives the probability of an outcome less than or equal to a given value. For a continuous distribution with boundaries, the probability of an outcome less than or equal to the lower bound is zero, and the probability of an outcome less than or equal to the upper bound is 1 (100%).

For Further Reference:

(Study Session 3, Module 9.1, LOS 9.c)

CFA® Program Curriculum, Volume 1, page 516

Question #31 of 120

Question ID: 1214583

Questions 31 through 42 relate to Economics. (18 minutes)

If there is an increase in the quantity of money at full employment, the long-run effects will *most likely* be:

A) an increase in the price level and a decrease in real GDP.

B) **an increase in the price level and no effect on real GDP.**

C) no effect on the price level and no effect on real GDP.

Explanation

An increase in the quantity of money at full employment will reduce interest rates in the short run, which would increase short-run aggregate demand above full employment. This causes wage demands to increase, which in turn reduces short-run aggregate supply. In the long run, real GDP reverts to its full-employment level with an increase in price level equal to the percentage increase in the quantity of money.

For Further Reference:

(Study Session 4, Module 14.3, LOS 14.i and Study Session 5, Module 16.2, LOS 16.k)

CFA® Program Curriculum, Volume 2, page 161

CFA® Program Curriculum, Volume 2, page 283

Question #32 of 120

Question ID: 1214586

A spot exchange rate is 8.6145 and the 1-year forward quotation is +0.25%. The 1-year forward quotation on a points basis is *closest* to:

A) 2.

B) 25.

C) **215.**

Explanation

Convert the percentage quote to a points quote as 0.0025 × 8.6145 = 0.0215, which is 215 points (each point is 0.0001).

For Further Reference:

(Study Session 5, Module 18.2, LOS 18.e)

CFA® Program Curriculum, Volume 2, page 421

Question #33 of 120

Question ID: 1214577

Which of the following *most likely* describes a loss that consumers suffer under an unregulated monopoly compared to a competitive market?

A) Monopolies produce less goods than a competitive market would.
B) Costs of production are higher with monopolies.
C) Monopolists charge the maximum price.

Explanation

A reduction in output and increase in price under monopoly decrease consumer surplus and welfare compared to perfect competition. A natural monopoly may have lower costs than several competitive suppliers. Monopolists charge the profit maximizing price, not the "maximum price."

For Further Reference:

(Study Session 4, Module 13.4, LOS 13.b)

CFA® Program Curriculum, Volume 2, page 64

Question #34 of 120

Question ID: 1214581

The national government has undertaken a plan to combat a recession that includes a fiscal stimulus package. The school of economic thought *most likely* to support this action is the:

A) Neoclassical.
B) Keynesian.
C) Monetarist.

Explanation

The Keynesian school of macroeconomics believes that increasing the money supply or increasing government spending (such as with a stimulus package) will increase real GDP and combat recession. Monetarists believe economic growth is best supported through a policy of steady and predictable money supply increases. The neoclassical school of thought believes that economic cycles will correct themselves through a rapid adjustment of the prices of key productive inputs that restores the economy to full employment.

For Further Reference:

(Study Session 4, Module 15.1, LOS 15.c)

CFA® Program Curriculum, Volume 2, page 211

Question #35 of 120

Question ID: 1214580

Under the production function approach to modeling economic growth, will growing an economy's amount of physical capital at a constant rate support long-term economic growth at that rate?

A) No, because physical capital depreciates over time.

B) Yes, because output is a positive function of both labor and capital.

C) **No, because physical capital exhibits diminishing marginal productivity.**

Explanation

A constant growth rate of physical capital will increase output over time, but because capital exhibits diminishing marginal productivity, the economy will grow at a rate that decreases over time.

For Further Reference:

(Study Session 4, Module 14.3, LOS 14.n, 14.o)

CFA® Program Curriculum, Volume 2, page 172

CFA® Program Curriculum, Volume 2, page 173

Question #36 of 120

Question ID: 1214587

Which of the following statements about the elasticities and absorption approaches to explaining the impact of exchange rate changes on trade deficits is *most accurate*?

A) Both the elasticities and absorption approaches consider trade and capital flows.

B) **Under the elasticities approach, currency depreciation will result in greater improvement in the trade deficit when either import or export demand becomes more elastic.**

C) Under the absorption approach, depreciation of the domestic currency will improve a trade deficit if it increases national expenditures relative to income.

Explanation

Under the elasticities approach, currency depreciation will result in greater improvement in the trade deficit when either import or export demand becomes more elastic. One shortcoming of the elasticities approach is that it considers trade flows and ignores capital flows. The absorption approach considers both. Under the absorption approach, depreciation of the domestic currency will improve the balance of trade if it increases domestic savings (i.e., increases national income relative to expenditures).

For Further Reference:

(Study Session 5, Module 18.3, LOS 18.j)

CFA® Program Curriculum, Volume 2, page 440

Question #37 of 120

Question ID: 1214578

Which model of pricing strategy under imperfect competition requires that firms be able to identify groups of customers that have different elasticities of demand?

A) Nash equilibrium.

B) **Price discrimination.**

C) Kinked demand curve.

Explanation

For a price discrimination strategy to increase a firm's profits, the firm must face a downward-sloping demand curve, have at least two identifiable groups of customers that have different price elasticities of demand, and be able to prevent these customers from reselling the product to each other.

For Further Reference:

(Study Session 4, Module 13.4, LOS 13.f)

CFA® Program Curriculum, Volume 2, page 64

Question #38 of 120

Question ID: 1214584

An advantage of a common market, relative to a customs union or a free trade area, is that a common market:

A) adopts a single currency.

B) **removes barriers to movement of labor and capital.**

C) establishes common institutions and economic policy.

Explanation

A free-trade area removes barriers to the flow of goods and services, while a common market removes barriers to movement of goods, services, labor, and capital.

For Further Reference:

(Study Session 5, Module 17.2, LOS 17.f)

CFA® Program Curriculum, Volume 2, page 358

Question #39 of 120

Question ID: 1214576

The demand function for a good is $Q_D = 108 – 6P_{GOOD} + 0.9\text{ Income} + 0.8\ P_{RELATED\ GOOD}$. If the price of the good is 15, income is 80, and the price of the related good is 20, the good's price elasticity of demand is *closest* to:

A) –0.4.

B) **–0.8.**

C) –1.2.

Explanation

At a price of 15, the quantity demanded is 108 − 6(15) + 0.9(80) + 0.8(20) = 106. The slope coefficient on the good's own price, ΔQ / ΔP, is –6. The price elasticity of demand at the current equilibrium price is –6(15 / 106) = –0.849.

For Further Reference:

(Study Session 4, Module 12.1, LOS 12.a)

CFA® Program Curriculum, Volume 2, page 9

Question #40 of 120

Question ID: 1214585

At a base period, the CPIs of the countries of Tuolumne (currency is the TOL) and Bodee (currency is the BDE) are both 100, and the exchange rate is 0.90 BDE/TOL. One year later, the exchange rate is 0.75 BDE/TOL, and the CPI has risen to 110 in Tuolumne and 105 in Bodee. The real exchange rate is *closest* to:

A) 0.72 BDE/TOL.

B) **0.79 BDE/TOL.**

C) 0.83 BDE/TOL.

Explanation

The real exchange rate is calculated as 0.75 BDE/TOL × 110/105 = 0.79 BDE/TOL.

For Further Reference:

(Study Session 5, Module 18.1, LOS 18.a)

CFA® Program Curriculum, Volume 2, page 397

Question #41 of 120

Question ID: 1214582

According to the quantity theory of money, the *most appropriate* means to combat inflation is to:

A) reduce the velocity of money.

B) **reduce the money supply.**

C) increase the excess reserves of banks.

Explanation

The quantity theory focuses on the quantity of money. The quantity theory states that velocity is not affected by monetary policy. Increasing banks' excess reserves would most likely lead to higher inflation.

For Further Reference:

(Study Session 5, Module 16.1, LOS 16.c)

CFA® Program Curriculum, Volume 2, page 264

Question #42 of 120

Question ID: 1214579

Compared to a competitive market result, a single-price monopoly will *most likely*:

A) adopt a marginal cost pricing strategy, which will decrease consumer surplus.

B) **result in a higher price, less consumer surplus, and more producer surplus.**

C) result in lower output, deadweight loss, and less producer and consumer surplus.

Explanation

A firm in a monopoly position will reduce output to where MC = MR, which will increase price, decrease consumer surplus, and increase producer surplus. A marginal cost pricing strategy refers to regulation which requires a firm to set price equal to marginal cost.

For Further Reference:

(Study Session 4, Module 13.4, LOS 13.d)

CFA® Program Curriculum, Volume 2, page 64

Question #43 of 120

Question ID: 1214641

Questions 43 through 60 relate to Financial Reporting and Analysis. (27 minutes)

Shelby Enterprises recently entered into a new $500 million revolving credit facility. The provisions of the facility require Shelby to repay the loan before any other debt can be retired. In addition, if the company's debt-to-capital ratio is higher than 1.0 or its equity falls below $2 billion, Shelby will be prohibited from paying any dividends. Shelby would *most likely* agree to these covenants because they reduce:

A) risk to bondholders.
B) **the company's interest cost.**
C) risk to shareholders.

Explanation

Shelby's management will agree to the lending covenants to lower the interest rate on the credit facility. The existing bondholders and shareholders may have more risk since their respective interests are subordinate to the credit facility.

For Further Reference:

(Study Session 8, Module 28.3, LOS 28.d and Study Session 15, Module 47.1, LOS 47.c)

CFA® Program Curriculum, Volume 3, page 504

CFA® Program Curriculum: Volume 5, page 305

Question #44 of 120

Question ID: 1214589

The objective of financial statements is *best* described as:

A) aiding in the evaluation of potential equity and debt investments.
B) **providing economic decision makers with useful information about a firm's financial performance and changes in financial position.**
C) providing shareholders detailed information about the performance and financial position of the firm.

Explanation

The objective of financial reporting is to ensure that the information in financial statements is useful to a wide range of users.

For Further Reference:

(Study Session 6, Module 20.1, LOS 20.a)

CFA® Program Curriculum, Volume 3, page 46

Question #45 of 120

Question ID: 1214595

A company understates year-end depreciation. As compared to the properly stated year-end results, what effect will this understatement have on the company's asset turnover ratio?

A) No impact.

B) **Decrease.**

C) Increase.

Explanation

Asset turnover equals sales / average total assets. Understating depreciation expense has no effect on sales. The lower depreciation will result in understatement of accumulated depreciation, so assets will be overstated. The higher level of assets will decrease the asset turnover ratio.

For Further Reference:

(Study Session 7, Modules 21.3, 24.2, LOS 21.d, 24.b)

CFA® Program Curriculum, Volume 3, page 84

CFA® Program Curriculum, Volume 3, page 256

Question #46 of 120

Question ID: 1214602

Deferred taxes must be recognized for undistributed earnings from an investment in an associate firm under:

A) **U.S. GAAP only.**

B) both IFRS and U.S. GAAP.

C) neither IFRS nor U.S. GAAP.

Explanation

Deferred taxes must be recognized for undistributed earnings from an investment in an associate firm under U.S. GAAP. Under IFRS, no deferred taxes are reported for undistributed earnings if the investor firm controls the sharing of profits and it is probable the temporary difference will not be reversed in the future.

For Further Reference:

(Study Session 8, Module 27.5, LOS 27.j)

CFA® Program Curriculum, Volume 3, page 477

Question #47 of 120

Question ID: 1214590

Thunderbird Company reported net income of $500 million and the company had 100 million common shares outstanding. In addition, Thunderbird had 5 million shares of convertible preferred and 10 million outstanding warrants during the year. Each preferred share pays a dividend of $4 per share and is convertible into three common shares. Each warrant is convertible into one common share at $25 per share. The company's stock traded at an average $50 per share. Thunderbird's diluted earnings per share for the year is *closest* to:

A) $4.00 per share.

B) **$4.20 per share.**

C) $4.80 per share.

Explanation

$$\text{Basic EPS} = \frac{\$500\text{ million}-\$20\text{ million}}{100\text{ million}} = \$4.80\text{ per share}$$

Preferred dividend = 5 million × $4 = $20 million

Shares created from conversion = 5 million shares × 3 shares = 15 million shares

$$\frac{\$20\text{ million}}{15\text{ million}} = \$1.33\text{, convertible preferred is dilutive}$$

$$\text{Shares created from warrant} = \left[\frac{\$50-\$25}{\$50}\right] \times 10\text{ million shares} = 5\text{ million shares}$$

Exercise price < average price, warrants are dilutive.

$$\text{Diluted EPS} = \frac{\$500\text{ million}}{100\text{ million}+15\text{ million}+5\text{ million}} = \$4.17\text{ per share}$$

For Further Reference:

(Study Session 7, Module 21.4, LOS 21.g, 21.h)

CFA® Program Curriculum, Volume 3, page 99

CFA® Program Curriculum, Volume 3, page 99

Question #48 of 120

Question ID: 1214592

Under IFRS, firms may report an investment in the equity securities of other companies at fair value through:

A) profit and loss only.

B) other comprehensive income only.

C) **either profit and loss, or other comprehensive income.**

Explanation

Under IFRS, firms have an irrevocable choice at the time of purchase to report equity securities at fair value through other comprehensive income. If they do not make this election, equity securities are reported at fair value through profit and loss.

For Further Reference:

(Study Session 7, Module 22.6, LOS 22.e)

CFA® Program Curriculum, Volume 3, page 133

Question #49 of 120

Question ID: 1214588

Declaration and payment of a dividend during the most recent accounting period would be shown on a company's statements of:

A) **cash flows and changes in equity.**

B) cash flows and comprehensive income.

C) changes in equity and comprehensive income.

Explanation

Declaration of a dividend will be recorded as a decrease in equity, and payment of a dividend will be shown on a company's cash flow statement, but neither affects income or comprehensive income.

For Further Reference:

(Study Session 6, Module 19.1, LOS 19.b)

CFA® Program Curriculum, Volume 3, page 13

Question #50 of 120

Question ID: 1214598

A firm ended the last period with inventory of $4.0 million and a LIFO reserve of $175,000. During the year, it made purchases of $2.0 million and reported sales of $5.5 million with a gross margin of 0.32. At the end of the year, it reported a LIFO reserve of $75,000. What is the value of the firm's cost of goods sold on a FIFO basis?

A) $3,640,000.
B) $3,740,000.
C) **$3,840,000.**

Explanation

With sales of $5.5 million and a gross margin of 0.32, COGS on a LIFO basis is $5.5 million × (1 − 0.32) = $3.74 million. To convert COGS to a FIFO basis, subtract the change in LIFO reserve during the year: $3,740,000 − ($75,000 − $175,000) = $3,840,000.

For Further Reference:

(Study Session 8, Module 25.3, LOS 25.f)

CFA® Program Curriculum, Volume 3, page 323

Question #51 of 120

Question ID: 1214599

Which of these intangible assets is *most likely* to be amortized?

A) Purchased patent that will expire in the current period.
B) **Purchased franchise right with a useful life of two years.**
C) Internally developed trademark with a useful life of 20 years.

Explanation

A purchased, identifiable intangible asset with a finite life is amortized over its useful life. Costs incurred to develop an intangible asset such as a trademark are expensed when incurred. A patent that expires in the current period will not provide future benefits and therefore should not be recognized as an asset.

For Further Reference:

(Study Session 8, Module 26.1, LOS 26.b)

SchweserNotes, Book 3 page 181

CFA® Program Curriculum, Volume 3, page 382

Question #52 of 120

Question ID: 1214593

During 20X1, Tusa Company sold machinery with an original cost of $100,000, and recognized a $15,000 gain from the sale. At the time of the sale, the accumulated depreciation of the machinery was $80,000. Ignoring taxes, the machinery sale will produce a:

A) $15,000 inflow from investing activities.
B) $20,000 inflow from operating activities.
C) $35,000 inflow from investing activities.

Explanation

Ignoring taxes, the cash flow for 20X1 consists of the sale proceeds. The sale proceeds equal $35,000, or the $20,000 book value ($100,000 cost – $80,000 accumulated depreciation) plus the $15,000 gain. The proceeds are reported as an inflow from investing activities.

For Further Reference:

(Study Session 7, Modules 23.1, 23.2, LOS 23.a, 23.f)

CFA® Program Curriculum, Volume 3, page 185

CFA® Program Curriculum, Volume 3, page 199

Question #53 of 120

Question ID: 1214594

David Chance, CFA, is analyzing Grow Corporation. Chance gathers the following information:

Net cash provided by operating activities	$3,500
Net cash used for fixed capital investments	$727
Cash paid for interest	$195
Income before tax	$4,400
Income tax expense	$1,540
Net income	$2,860

Grow's free cash flow to the firm (FCFF) is *closest* to:

A) $2,260.
B) $2,640.
C) $2,900.

Explanation

$$\text{FCFF} = \text{CFO} + \text{Int}(1 - \text{tax rate}) - \text{capital expenditures}$$

$$\text{FCFF} = 3{,}500 + \left[195 \times \left(1 - \left(\frac{1{,}540}{4{,}400}\right)\right)\right] - 727 = 2{,}899.75 \approx 2{,}900$$

For Further Reference:

(Study Session 7, Module 23.4, LOS 23.i)

CFA® Program Curriculum, Volume 3, page 222

Question #54 of 120

Question ID: 1214597

For the last few years, firms in an expanding industry have found it more difficult to keep up with consumer demand despite steadily increasing inventory levels. The Consumer Price Index (CPI) has been at a level of 1050, 1060, and 1087 in the last three years. Given this situation, a firm in this industry that seeks to report higher net income would *most likely* prefer which inventory accounting method?

A) LIFO.
B) **FIFO.**
C) Average cost.

Explanation

In a period of rising prices and rising inventory levels, FIFO results in the highest net income (lowest COGS).

For Further Reference:

(Study Session 8, Modules 25.1, 25.2, LOS 25.c, 25.d)

CFA® Program Curriculum, Volume 3, page 316

CFA® Program Curriculum, Volume 3, page 320

Question #55 of 120

Question ID: 1214603

Jo Evans, CFA, analyzes the financial statements of Shubert Company and writes, "Shubert's earnings, while sustainable, provide an inadequate return to shareholders." Evans has expressed a concern with Shubert's:

A) **quality of reported results only.**
B) financial reporting quality only.
C) quality of reported results and financial reporting quality.

Explanation

Adequacy and sustainability of earnings are issues related to quality of reported results. Financial reporting quality refers to issues related to the relevance and faithful representation of financial reports.

For Further Reference:

(Study Session 9, Module 29.1, LOS 29.a)

CFA® Program Curriculum, Volume 3, page 540

Question #56 of 120

Question ID: 1214601

Selected information on Reckner Company's income taxes appears in the following table:

	20X1	20X2	20X3

Taxes payable	250	500	500
Deferred tax assets	200	300	200
Deferred tax liabilities	200	300	400

Compared to 20X2, Reckner's 20X3 income tax expense:

- **A) increased.**
- **B)** decreased.
- **C)** remained the same.

Explanation

Income tax expense = taxes payable + ΔDTL – ΔDTA. Income tax expense in 20X2 was 500 + (300 − 200) − (300 − 200) = 500. Income tax expense in 20X3 was 500 + (400 − 300) − (200 − 300) = 700. Income tax expense increased by 200.

For Further Reference:

(Study Session 8, Module 27.3, LOS 27.d)

CFA® Program Curriculum, Volume 3, page 454

Question #57 of 120

Question ID: 1214600

A decrease in accumulated depreciation is *most likely* to result from:

- **A) selling or disposing of a long-lived asset.**
- **B)** increasing the salvage value of a long-lived asset.
- **C)** decreasing the estimated useful life of a long-lived asset.

Explanation

When a long-lived asset is sold or otherwise disposed of, its original cost and accumulated depreciation are removed from the balance sheet. Changing the estimated salvage value or useful life of a long-lived asset will change depreciation expense in the subsequent periods but does not affect accumulated depreciation.

For Further Reference:

(Study Session 8, Module 26.3, LOS 26.j)

CFA® Program Curriculum, Volume 3, page 415

Question #58 of 120

Question ID: 1214591

Magnus Aerospace produces and sells aircraft and has approximately a 2-year operating cycle. Magnus's liabilities include commercial paper due in 270 days, a bank note due in one year, and bonds that will mature in 18 months. Magnus should *most appropriately* classify as current liabilities:

- **A) all of these liabilities.**
- **B)** only the commercial paper.
- **C)** the commercial paper and the bank note.

Explanation

Current liabilities are obligations due within one year or the company's operating cycle, whichever is longer. With an operating cycle of two years, Magnus should classify as current any liabilities that must be settled in less than two years.

For Further Reference:

(Study Session 7, Module 22.3, LOS 22.d)

CFA® Program Curriculum, Volume 3, page 131

Question #59 of 120

Question ID: 1214604

An analyst gathers the following selected financial information on Quip Corp.

Partial financials for 20X8	Quip Corp
Sales	$350,000
Cost of goods sold	270,000
Net income	35,000
Current assets	165,000
Current liabilities	130,000
20X8 LIFO reserve	30,000
20X7 LIFO reserve	20,000

To compare Quip and its competitors, an analyst makes the necessary adjustments to restate Quip's financial statements to reflect the FIFO inventory accounting method. Quip's adjusted gross profit margin is *closest* to:

A) 20%.
B) 23%.
C) **26%.**

Explanation

adjusted cost of goods sold Quip = 270,000 − (30,000 − 20,000) = 260,000

adjusted gross profit margin Quip = (350,000 − 260,000) / 350,000 = 0.257 ≈ 26%

For Further Reference:

(Study Session 9, Module 30.2, LOS 30.e)

CFA® Program Curriculum, Volume 3, page 627

Question #60 of 120

Question ID: 1214596

Hazel Edwards, CFA, is analyzing Collins Footwear, Inc. and obtains the following data for the company's major geographic segments:

	Asia	Europe	North America
Sales (U.S. dollars)	$200 million	$300 million	$500 million

Net profit margin	4.5%	3.0%	1.5%
Asset turnover	1.5×	2.5×	4.0×

Based on these data, Edwards should conclude that Collins's:

A) smallest segment by assets is Asia.

B) **most profitable segment by return on assets is Europe.**

C) most profitable segment by net income is North America.

Explanation

Return on assets is 3.0 × 2.5 = 7.5% in Europe, 4.5 × 1.5 = 6.75% in Asia, and 1.5 × 4.0 = 6.0% in North America.

Assets are $200 million / 1.5 = $133.3 million in Asia, $500 million / 4.0 = $125 million in North America, and $300 million / 2.5 = $120 million in Europe.

Net income is 4.5% × $200 million = $9 million in Asia, 3.0% × $300 million = $9 million in Europe, and 1.5% × $500 million = $7.5 million in North America.

For Further Reference:

(Study Session 7, Module 24.5, LOS 24.f)

CFA® Program Curriculum, Volume 3, page 292

Question #61 of 120

Question ID: 1214608

Questions 61 through 72 relate to Corporate Finance. (18 minutes)

Companies that can only raise a limited amount of funds to invest in projects must use:

A) **capital rationing.**

B) project sequencing.

C) capital preservation.

Explanation

Capital rationing, or prioritizing projects to maximize the increase in company value, is necessary when the firm only has a limited amount of funds to invest. Project sequencing involves more than one project sequenced over time, and an unfavorable outcome for one project can cause cancellation of the next project in the sequence. Capital preservation is an investment return objective that involves an investment return at least equal to the inflation rate with minimal chance of loss.

For Further Reference:

(Study Session 10, Module 32.1, LOS 32.c)

CFA® Program Curriculum, Volume 4, page 51

Question #62 of 120

Question ID: 1214605

Special resolutions that require a supermajority of shareholder votes may be addressed:

A) only at the annual general meeting.

B) only at an extraordinary general meeting.

C) at either the annual general meeting or an extraordinary general meeting.

Explanation

Special resolutions may be voted on at the annual general meeting or at an extraordinary general meeting that is called specifically to address them.

For Further Reference:

(Study Session 10, Module 31.1, LOS 31.e)

CFA® Program Curriculum, Volume 4, page 14

Question #63 of 120

Question ID: 1214613

Other things equal, a company's operating breakeven level of sales is *most likely* to increase when:

A) the tax rate is increased.

B) its scale of operations is increased.

C) fixed interest charges are increased.

Explanation

Operating breakeven is calculated as fixed operating costs divided by price minus variable costs per unit. Tax rates and interest charges do not affect the operating breakeven quantity of sales.

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.e)

CFA® Program Curriculum, Volume 4, page 143

Question #64 of 120

Question ID: 1214609

Inverness Corporation is considering investing in one of two mutually exclusive projects. The firm's cost of capital is 15%. Project One's NPV profile crosses the vertical axis at $1.8 million and crosses the horizontal axis at 25%. Project Two's NPV profile crosses the vertical axis at $1.2 million and crosses the horizontal axis at 33%. For the two projects, the crossover rate is 18%. Inverness should:

A) accept Project One.

B) accept Project Two.

C) reject both projects.

Explanation

The NPV of Project One is higher than the NPV of Project Two at any discount rate less than the crossover rate, and the NPVs of both projects are positive at the firm's cost of capital. Because Project One and Project Two are mutually exclusive, the firm should select Project One because it has the higher NPV at a discount rate of 15%.

For Further Reference:

(Study Session 10, Module 32.2, LOS 32.e)

CFA® Program Curriculum, Volume 4, page 58

Question #65 of 120

Question ID: 1214607

Which of the following factors should an analyst *most likely* consider favorable for shareholders' interests?

A) Non-executive directors.
B) Anti-takeover provisions.
C) Dual-class share structures.

Explanation

Non-executive, or external, directors are typically viewed as more likely to act in shareholders' interests than executive, or internal, directors. Dual-class structures may allow a small group of shareholders, such as company founders and their heirs, to exercise control disproportionate to their ownership stake. Anti-takeover provisions limit shareholders' ability to bring about changes in management.

For Further Reference:

(Study Session 10, Module 31.2, LOS 31.i)

CFA® Program Curriculum, Volume 4, page 31

Question #66 of 120

Question ID: 1214611

The cost of preferred equity in a company's capital structure is based on the relationship between a preferred share's:

A) dividend and par value.
B) dividend and market price.
C) dividend yield and expected growth.

Explanation

The cost or preferred equity is calculated by dividing the preferred dividend by the market price of a preferred share. Regular preferred dividends do not grow over time.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.g)

CFA® Program Curriculum, Volume 4, page 86

Question #67 of 120

Question ID: 1214612

To choose the weights for a firm's weighted average cost of capital (WACC), an analyst should *most appropriately* use the:

A) firm’s current debt and equity weights based on market value.
B) firm’s stated target capital structure even though recent fund raising has diverged slightly from the target weights.
C) average debt and equity weights based on market value of the firm’s competitors.

Explanation

The analyst should use the target capital structure weighting if possible (i.e., if the firm has stated its target weights). If the targets are unknown, the analyst must estimate the weights.

For Further Reference:

(Study Session 10, Module 33.1, LOS 33.c)

CFA® Program Curriculum, Volume 4, page 80

Question #68 of 120

Question ID: 1214615

Drags on liquidity are *most likely* to include:

A) inventory obsolescence.
B) paying vendors too soon.
C) suppliers requiring payment sooner.

Explanation

Drags on liquidity either delay or reduce cash inflows. Obsolete inventory can be expected to sell more slowly and often will require write-downs of inventory value. Paying vendors sooner than is optimal and shorter payment periods for suppliers accelerate cash outflows and are referred to as pulls on liquidity.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.a)

CFA® Program Curriculum, Volume 4, page 157

Question #69 of 120

Question ID: 1214614

A company with which of the following leverage measures would exhibit the least sensitivity of earnings per share to a change in sales?

	Degree of operating leverage	Degree of financial leverage
A)	**1.5**	**3.0**
B)	2.0	2.5
C)	2.5	2.0

Explanation

The degree of total leverage (DTL) measures the sensitivity of earnings per share to a change in sales. With a DOL of 1.5 and a DFL of 3.0, the DTL is 1.5 × 3.0 = 4.5. With DOL of 2.0 and DFL of 2.5, or with DOL of 2.5 and DFL of 2.0, the DTL would be 2.0 × 2.5 = 5.0.

For Further Reference:

(Study Session 11, Module 34.1, LOS 34.b)

CFA® Program Curriculum, Volume 4, page 128

Question #70 of 120

Question ID: 1214616

Shawn Wright, CFA, is evaluating the short-term investment policy for Hegeman Industries. Wright should *most likely* conclude that Hegeman's investment policy is:

A) inappropriate if it restricts the types of securities that can be held.

B) appropriate if it lists specific issuers from which Hegeman may purchase securities.

C) **appropriate if it limits the proportion of the total portfolio that can be held in various types of issues.**

Explanation

An appropriate short-term investment policy statement should include limitations on the proportions of the total short-term securities portfolio that can be invested in the various types of permitted securities. The policy statement should also include limitations on the types of securities that can be held. It would be overly restrictive, however, to include a specific listing of issuers from which securities could be purchased.

For Further Reference:

(Study Session 11, Module 35.1, LOS 35.e)

CFA® Program Curriculum, Volume 4, page 169

Question #71 of 120

Question ID: 1214606

A company's corporate governance procedures, with respect to stakeholder management, are *best* described as part of its:

A) legal infrastructure.

B) contractual infrastructure.

C) **organizational infrastructure.**

Explanation

Organizational infrastructure refers to a company's corporate governance procedures and includes its internal systems and practices that address how it manages its stakeholder relationships. Legal infrastructure refers to laws that govern stakeholder relationships. Contractual infrastructure refers to contracts between a company and its stakeholders.

For Further Reference:

(Study Session 10, Module 31.1, LOS 31.d)

CFA® Program Curriculum, Volume 4, page 14

Question #72 of 120

Question ID: 1214610

A company announces a new project that an analyst believes will require an investment of $5 per share this year and will increase after-tax cash flow by $1 per share in each of the next eight years. Assuming this project is new information and does not have any signaling effects, what should the analyst estimate as the impact of acceptance of the project on the value of the company's shares?

A) An increase of $3.

B) **Either an increase or a decrease.**

C) An increase of less than $3.

Explanation

Other things equal, the new project should change the share value by the net present value (NPV) of the new project per share. Although the project has a positive internal rate of return (IRR) (total inflows are greater than the initial outlay), no information is given about the company's cost of capital. If the company's cost of capital is less than the project's IRR, the project's NPV will be positive and share value will be increased. If the cost of capital is greater than the project's IRR, the project will have a negative NPV and decrease share value.

For Further Reference:

(Study Session 10, Module 32.2, LOS 32.g)

CFA® Program Curriculum, Volume 4, page 66

Question #73 of 120

Question ID: 1214623

Questions 73 through 86 relate to Equity Investments. (21 minutes)

The *most likely* reason that a company's book value of equity and the market value of its equity securities may be different is that:

A) market value of equity securities includes preferred stock, but the book value of equity does not.

B) the issuance of new equity will increase the book value of equity, but will not affect the share price.

C) **the market value of equity securities reflects investor expectations about future performance, but book value of equity does not.**

Explanation

The market value of equity securities is affected by investor expectations about future firm performance, while the book value of equity is based on the firm's past performance. The book value of equity includes both common and preferred equity. The issuance of new equity will increase the market value of equity, even if the share price is unaffected.

For Further Reference:

(Study Session 13, Module 39.2, LOS 39.g)

CFA® Program Curriculum, Volume 4, page 373

Question #74 of 120

Question ID: 1214626

A firm attempts to gain market share from its competitors by improving its manufacturing efficiency so that it can increase output and reduce the price of its product. This firm's competitive strategy is *most accurately* described as:

A) an offensive differentiation strategy.

B) **an offensive cost leadership strategy.**

C) a defensive cost leadership strategy.

Explanation

Cost leadership is a competitive strategy in which a firm attempts to become the lowest-cost producer in the industry. This will enable the firm to offer lower prices than its competitors. A differentiation strategy involves making a product distinctive compared to competing products so that customers will be willing to pay a premium price. A competitive strategy is described as defensive if it is used to maintain a firm's market share and offensive if it is used to gain market share.

For Further Reference:

(Study Session 13, Module 40.2, LOS 40.k)

CFA® Program Curriculum, Volume 4, page 429

Question #75 of 120

Question ID: 1214627

Yong Kim, CFA, buys a preferred stock that has a 6% dividend yield (defined as the ratio of the preferred dividend to the market price of the preferred stock). One year later, Kim sells the stock when it is selling at a 5% dividend yield. The preferred stock pays a fixed annual dividend, which Kim received right before selling. What rate of return did Kim realize on his investment?

A) 14%.
B) 20%.
C) **26%.**

Explanation

The dividend can be of any size. Suppose it is $1.00.

The purchase price is 1.00 / 0.06 = 16.667.

The sale price is 1.00 / 0.05 = 20.

Kim pays 16.667 and receives 20.00 plus a 1.00 dividend one year later. The rate of return is $[(20 + 1)/16.667] − 1 = 26\%$.

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.f)

CFA® Program Curriculum, Volume 4, page 456

Question #76 of 120

Question ID: 1214624

A U.S. investor purchases ADRs of a Japanese company, while a Japanese investor purchases the same value of the company's common stock. Compared to the Japanese investor, the U.S. investor will *most likely*:

A) face the same risk.
B) **realize different returns.**
C) benefit from greater transparency.

Explanation

The return to the U.S. investor is affected by the return on the shares in Japanese yen and by the dollar/yen exchange rate. The U.S. investor therefore faces additional currency risk, which will most likely result in returns that differ from those a Japanese investor would realize. ADRs do not necessarily offer greater transparency to foreign investors than that which is available to domestic investors.

For Further Reference:

(Study Session 13, Module 39.2, LOS 39.d)

CFA® Program Curriculum, Volume 4, page 365

Question #77 of 120

Question ID: 1214628

Larry Rile is evaluating the investment merits of Bing Corp., a successful motorcycle manufacturer. Rile is forecasting a dividend in year 1 of $1.50 per share, a dividend in year 2 of $3.00 per share, and a dividend in year 3 of $4.50 per share. After year 3, Rile expects dividends to grow at the rate of 6% per year. Rile calculates a beta of 1.3 for Bing. Rile expects the S&P 500 index to return 8%. The U.S. Treasury bill is yielding 2%. Using the multistage dividend discount model, Bing's intrinsic value is *closest* to:

A) $92 per share.

B) **$102 per share.**

C) $112 per share.

Explanation

D1 = 1.50

D2 = 3.00

D3 = 4.50

g = 0.06

required return = RFR + β(R_m − RFR)

required return = 0.02 + 1.3(0.08 − 0.02) = 0.098

$$\mathrm{P}_3=\frac{4.50\times 1.06}{0.098-0.06}=\frac{4.77}{0.038}=125.53$$

$$\mathrm{V}=\frac{1.50}{1.098}+\frac{3.00}{(1.098)^2}+\frac{4.50+125.53}{(1.098)^3}=102.08$$

For Further Reference:

(Study Session 13, Module 41.2, LOS 41.g)

CFA® Program Curriculum, Volume 4, page 459

Question #78 of 120

Question ID: 1214618

A securities exchange is structured as a call market. On that exchange, a stock would trade at:

A) any time the market is open.

B) **one negotiated price that clears the market.**

C) prices set by auction or by dealer bid-ask quotes.

Explanation

In a call market, stocks trade at specific times at one price that clears the market for the stock. In continuous markets, trades occur at any time the market is open at prices set by auction or by dealer quotes.

For Further Reference:

(Study Session 12, Module 36.3, LOS 36.j)

CFA® Program Curriculum, Volume 4, page 256

Question #79 of 120

Question ID: 1214629

Enterprise value is *most accurately* interpreted as the:

A) **cost to take over a firm.**
B) fair market value of a firm’s equity.
C) market value of a firm’s equity plus the market value of its debt.

Explanation

Enterprise value equals the market value of a firm's common stock and debt minus its cash and short-term securities. Enterprise value is interpreted as the amount that would be needed to take over a firm.

For Further Reference:

(Study Session 13, Module 41.3, LOS 41.k)

CFA® Program Curriculum, Volume 4, page 477

Question #80 of 120

Question ID: 1214625

James Martindale, CFA, manages a small mutual fund specializing in defensive stocks. For this fund, Martindale will buy stocks with:

A) high beta.
B) **low systematic risk.**
C) low price-to-earnings ratios.

Explanation

Defensive stocks (with low betas and low systematic risk) are less sensitive to economic cycles. A high beta stock is a cyclical stock. Stocks with low P/E ratios are value stocks.

For Further Reference:

(Study Session 13, Module 40.1, LOS 40.c)

CFA® Program Curriculum, Volume 4, page 390

Question #81 of 120

Question ID: 1214622

In an informationally efficient capital market, a portfolio manager who follows an active investment strategy is *most likely* to:

A) outperform a passive index strategy on average over time.
B) **underperform a passive index strategy on average over time.**
C) earn the same net return over time as a passive index strategy.

Explanation

In an informationally efficient market, prices are unbiased estimates of securities' intrinsic values. Therefore, active management cannot consistently generate positive risk-adjusted returns compared to a passive index strategy. Because transactions costs reduce its net return, an active strategy underperforms a passive index strategy on average over time.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.a)

CFA® Program Curriculum, Volume 4, page 319

Question #82 of 120

Question ID: 1214619

Which of the following limit orders is *least likely* to be filled?

A) Inside-the-market limit sell order.

B) Behind-the-market limit buy order.

C) Aggressively priced limit buy order.

Explanation

"Behind the market" refers to a buy order with a limit price less than the best bid, or a sell order with a limit price higher than the best ask. A limit order that is behind the market (i.e., an order to buy for less than the market price or sell for more than the market price) will not be filled unless the market price moves to the order's limit price. "Inside the market" refers to orders with limit prices between the best bid and best ask. "Aggressively priced" refers to a buy order with limit prices higher than the best ask or a sell order with a limit price lower than the best bid. Aggressively priced limit orders are most likely to be filled immediately.

For Further Reference:

(Study Session 12, Module 36.3, LOS 36.h)

CFA® Program Curriculum, Volume 4, page 246

Question #83 of 120

Question ID: 1214620

Creating a bond market index is more difficult than constructing a stock market index due to:

A) lack of continuous trade data for bonds.

B) lower price volatility of bonds versus stocks.

C) a narrower universe of bonds versus stocks.

Explanation

It is difficult to price individual bond issues in an index because continuous trade data may not exist for some bonds. In addition, it is challenging to create a bond market index because the bond universe is much broader, and the price volatility of a bond (i.e., its duration) changes over time as the bond approaches maturity.

For Further Reference:

(Study Session 12, Module 37.2, LOS 37.i)

CFA® Program Curriculum, Volume 4, page 300

Question #84 of 120

Question ID: 1214621

A given percentage change in one of the 30 stocks in the Dow Jones Industrial Average (DJIA) will have the greatest impact on the DJIA for which index stock?

A) The one whose total equity has the highest market value.

B) The one whose stock trades at the highest dollar price per share.

C) The one having the greatest amount of equity in its capital structure.

Explanation

The Dow Jones Industrial Average is a price-weighted index. If each of its stocks changes by the same percentage, the stock with the largest impact on the index will be the one with the highest price per share.

For Further Reference:

(Study Session 12, Modules 37.1, 37.2, LOS 37.d, 37.k)

CFA® Program Curriculum, Volume 4, page 285

CFA® Program Curriculum, Volume 4, page 305

Question #85 of 120

Question ID: 1214630

Jim Boo, CFA, is analyzing Justin Corp., a maker of home appliances. Boo's research provides the following facts:

- Justin's stock price is $60 per share.
- Expected growth rate of dividends is 5%.
- Expected retention ratio is 60%.
- Required rate of return is 10%.

Justin's expected price to earnings ratio (P_0/E_1) is *closest* to:

A) 8.0x.

B) 10.0x.

C) 12.0x.

Explanation

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{k-g} = \frac{1-0.6}{0.1-0.05} = 8$$

For Further Reference:

(Study Session 13, Module 41.3, LOS 41.j)

SchweserNotes, Book 4 page 215

CFA® Program Curriculum, Volume 4, page 468

Question #86 of 120

Question ID: 1214617

An investor purchased a stock for $60 a share using margin from his broker. If the initial margin requirement is 40%, and the maintenance margin requirement is 20%, a margin call will initially be triggered below a share price of:

- **A)** $30.
- **B)** **$45.**
- **C)** $48.

Explanation

$$\text{Margin call price}= \text{P}_0\left(\frac{1-\text{initial margin\%}}{1-\text{maintenance margin\%}}\right)=\$60\left(\frac{1-0.40}{1-0.20}\right)=\$60\,(0.75)=\$45$$

For Further Reference:

(Study Session 12, Module 36.2, LOS 36.f)

CFA® Program Curriculum, Volume 4, page 243

Question #87 of 120

Question ID: 1214632

Questions 87 through 99 relate to Fixed Income. (19.5 minutes)

Which of the following sources of short-term funding for banks is *most likely* to have the lowest interest cost?

- **A)** Interbank funds.
- **B)** **Checking deposits.**
- **C)** Central bank funds.

Explanation

Typically, banks do not pay interest on customer checking account deposits. A bank that borrows in either the central bank funds market or the interbank market incurs interest costs.

For Further Reference:

(Study Session 14, Module 43.2, LOS 43.i)

CFA® Program Curriculum: Volume 5, page 90

Question #88 of 120

Question ID: 1214638

The current price of a $1,000 par value, 6-year, 4.2% semiannual coupon bond is $958.97. The bond's price value of a basis point is *closest* to:

- **A)** **$0.50.**
- **B)** $4.20.
- **C)** $5.01.

Explanation

First we compute the yield to maturity of the bond. PV = –$958.97, FV = $1,000, PMT = $21, N = 12, CPT I/Y = 2.5%, multiply by 2 since it is a semiannual bond to get an annualized yield to maturity of 5.0%. Now compute the price of the bond at using yield one basis point higher, or 5.01%. FV = $1,000, PMT = 21, N = 12, I/Y = (5.01 / 2 =) 2.505, CPT PV = –$958.47. The price changes from $958.97 to $958.47, or $0.50.

For Further Reference:

(Study Session 15, Module 46.2, LOS 46.g)

CFA® Program Curriculum: Volume 5, page 267

Question #89 of 120

Question ID: 1214639

Which of the following bonds is *most likely* to have the greatest convexity at a yield to maturity of 3.5%?

A) 2-year, 3% coupon bond.
B) 5-year, 4% coupon bond.
C) **10-year, zero-coupon bond.**

Explanation

Other things equal, convexity is greater for bonds with longer maturities and lower coupon rates.

For Further Reference:

(Study Session 15, Module 46.3, LOS 46.h)

CFA® Program Curriculum: Volume 5, page 269

Question #90 of 120

Question ID: 1214643

Yield spreads are *most likely* to widen in a market environment that exhibits:

A) **slowing economic growth.**
B) high GDP growth rates.
C) lower-than-normal supply of new bond issuance.

Explanation

Slowing economic growth reduces firm profitability and creditworthiness and causes yield spreads to widen. Higher-than-normal supply of new bonds can cause yield spreads to widen. Yield spreads tend to narrow during expansions when GDP growth rates are high.

For Further Reference:

(Study Session 15, Module 47.2, LOS 47.i)

CFA® Program Curriculum: Volume 5, page 338

Question #91 of 120

Question ID: 1214633

A 10-year, 5% bond is issued at a price to yield 5.2%. Three months after issuance, the yield on this bond has decreased by 100 basis points. The price of this bond at issuance and three months later is:

A) above par at issuance, but below par three months later.

B) **below par at issuance, but above par three months later.**

C) below par at issuance, and below par three months later.

Explanation

A bond issued at a yield higher than its coupon will be priced below par, or at a discount. Three months later, the yield has declined to 4.2% and the bond will trade at a premium to par, reflecting the fact that the coupon is now higher than the yield.

For Further Reference:

(Study Session 14, Module 44.3, LOS 44.b)

CFA® Program Curriculum: Volume 5, page 113

Question #92 of 120

Question ID: 1214635

Kathy Hurst, CFA, is valuing a 4-year zero coupon security and has acquired the following information:

1-year spot rate	6.0%
4-year spot rate	7.5%
1-year forward rate 1 year from now	7.3%
1-year forward rate 3 years from now	8.9%

The 1-year forward rate 2 years from now is *closest* to:

A) 7.3%.

B) **7.8%.**

C) 8.0%.

Explanation

$$\begin{aligned} (1+\mathrm{S}_4)^4 &= (1+\mathrm{S}_1)(1+1\mathrm{y}1\mathrm{y})(1+2\mathrm{y}1\mathrm{y})(1+3\mathrm{y}1\mathrm{y}) \\ (1.075)^4 &= (1.06)(1.073)(1+2\mathrm{y}1\mathrm{y})(1.089) \\ 1+2\mathrm{y}1\mathrm{y} &= 0.078 \end{aligned}$$

For Further Reference:

(Study Session 14, Module 44.4, LOS 44.j)

CFA® Program Curriculum: Volume 5, page 143

Question #93 of 120

Question ID: 1214640

Effective duration is the *most appropriate* measure of interest rate risk when:

A) a bond's credit rating may change significantly.

B) the investor is seeking to minimize a duration gap.

C) **the path of interest rate changes affects a bond's cash flows.**

Explanation

For bonds with future cash flows that depend on the path of interest rate changes, such as those with embedded options, effective duration is the most appropriate measure of interest rate risk. A duration gap relative to an investor's horizon is measured using Macaulay duration. The effect of rating changes may be estimated using credit duration.

For Further Reference:

(Study Session 17, Modules 46.1, 46.3, LOS 46.b, 46.c, 46.k)

CFA® Program Curriculum: Volume 5, page 248

CFA® Program Curriculum: Volume 5, page 255

CFA® Program Curriculum: Volume 5, page 279

Question #94 of 120

Question ID: 1214644

A bond indenture states that the source of funds for repayment will be tolls paid by drivers using a highway constructed with the bond proceeds. This bond is *most likely* a:

A) secured bond.

B) **revenue bond.**

C) quasi-government bond.

Explanation

Revenue bonds are municipal bonds that will be repaid from revenues generated by a specific project such as a toll road. Secured bonds have specific assets pledged as collateral. Quasi-government entities are agencies created by sovereign governments.

For Further Reference:

(Study Session 14, Module 42.1, LOS 42.a and Study Session 15, Module 47.2, LOS 47.j)

CFA® Program Curriculum: Volume 5, page 7

CFA® Program Curriculum: Volume 5, page 347

Question #95 of 120

Question ID: 1214636

A collateralized mortgage obligation with agency RMBS as the collateral is *least likely* to be created to offer securities with less:

A) **default risk than the underlying RMBS.**

B) extension risk than the underlying RMBS.

C) prepayment risk than the underlying RMBS.

Explanation

Agency CMOs are created to reapportion prepayment risk, which includes extension and contraction risk. Agency CMOs have little or no default risk because they are backed by the government or by GSEs.

For Further Reference:

(Study Session 14, Modules 45.1, 45.2, LOS 45.e)

CFA® Program Curriculum: Volume 5, page 194

Question #96 of 120

Question ID: 1214642

Bond X carries a rating of BBB-/Baa3. Bond Y has a rating of B/B2. Both bonds mature in 10 years. Which bond's value would be *most* affected by a ratings downgrade, and which bond has the higher default risk?

- **A) Bond X would be more affected by a ratings downgrade, but Bond Y has higher default risk.**
- **B)** Bond Y would be more affected by a ratings downgrade, but Bond X has higher default risk.
- **C)** Bond X has higher default risk, but both bonds would experience similar effects of a ratings downgrade.

Explanation

Bond X carries the lowest investment grade rating. If it is downgraded, it will fall to a speculative rating, and many investors will be restricted from owning it. A downgrade would therefore have a more significant impact on Bond X. Bond Y carries a speculative rating that implies more risk of default than the higher rated Bond X.

For Further Reference:

(Study Session 14, Module 43.1, LOS 43.a and Study Session 15, Module 47.1, LOS 47.a)

CFA® Program Curriculum: Volume 5, page 56

CFA® Program Curriculum: Volume 5, page 302

Question #97 of 120

Question ID: 1214631

The bid-ask spread for a bond *most likely* conveys information about:

- **A) its liquidity, but not its credit quality.**
- **B)** its credit quality, but not its liquidity.
- **C)** both its liquidity and its credit quality.

Explanation

Bond dealers' bid-ask spreads depend primarily on the liquidity of an issue. Spreads are narrower for highly liquid issues and wider for less liquid issues. Credit quality and liquidity are both reflected in yield spreads.

For Further Reference:

(Study Session 14, Module 43.1, LOS 43.d)

CFA® Program Curriculum: Volume 5, page 70

Question #98 of 120

Question ID: 1214637

An investor buys an annual pay, 4% coupon bond for 102. The trade will settle immediately after the annual coupon payment and the bond has five years left to maturity. The investor sells the bond two years later for 101.5. The investor's holding period return on the bond includes:

A) a capital loss.

B) **a capital gain.**

C) neither a capital loss nor gain.

Explanation

The yield to maturity at the time of purchase is:

N = 5; PV = –102; PMT = 4; FV = 100; CPT I/Y = 3.5563%.

The bond's constant-yield price two years later is:

N = 3; I/Y = 3.5563; PMT = 4; FV = 100; CPT PV = –101.24.

Because the price is greater than the constant-yield price, the investor has a capital gain on the sale.

For Further Reference:

(Study Session 15, Module 46.1, LOS 46.a)

CFA® Program Curriculum: Volume 5, page 241

Question #99 of 120

Question ID: 1214634

Gerald Snow is a bond manager for Long Vision Investments. Snow is evaluating potential arbitrage opportunities. He has the following list of bonds:

- Bond X is a 1-year zero coupon bond selling at 950.
- Bond Y is a 2-year zero coupon bond selling at 850.
- Bond Z is a 2-year bond with an annual coupon of 8%.

All three bonds have a par value of $1,000. If no arbitrage opportunity exists, the price of Bond Z is *closest* to:

A) $975.

B) **$995.**

C) $1,015.

Explanation

Bond X = 1000 / 950 = 1.0526; Bond Y = 1000 / 850 = 1.1765

80 / (1.0526) = 76; 1080 / (1.1765) = 917.98

Bond Z = 76 + 917.98 = 993.98 ≈ 995

The arbitrage-free valuation approach applies time-appropriate spot interest rates to each cash flow of the bond.

For Further Reference:

(Study Session 14, Module 44.2, LOS 44.c)

CFA® Program Curriculum: Volume 5, page 117

Question #100 of 120

Question ID: 1214649

Questions 100 through 106 relate to Derivatives. (10.5 minutes)

The value of a European put option at expiration is *most likely* to be increased by:

A) a higher exercise price.
B) a lower risk-free interest rate.
C) higher volatility of the underlying asset price.

Explanation

The value of an option at expiration is the greater of zero or its exercise value. A higher exercise price increases the exercise value of a put option because it gives the holder the right to sell the underlying asset for a higher price. The risk-free interest rate and volatility of the underlying asset price only affect the time value of options, which is zero at expiration.

For Further Reference:

(Study Session 16, Modules 48.2, 49.3, LOS 48.d, 49.k)

CFA® Program Curriculum, Volume 5, page 406

CFA® Program Curriculum, Volume 5, page 482

Question #101 of 120

Question ID: 1214645

A characteristic that distinguishes derivatives from ETFs is that derivatives:

A) transform the performance of their underlying assets.
B) are contingent claims on the performance of their underlying assets.
C) derive their performance from the performance of their underlying assets.

Explanation

Derivatives typically transform the performance of their underlying assets, in contrast with passing the performance of the underlying assets through to the investor in the manner of mutual funds or ETFs. Mutual funds and ETFs, as well as derivatives, derive their performance from the performance of their underlying assets. Derivatives may be contingent claims or forward commitments.

For Further Reference:

(Study Session 16, Module 48.1, LOS 48.a, 48.b)

CFA® Program Curriculum, Volume 5, page 386

CFA® Program Curriculum, Volume 5, page 387

Question #102 of 120

Question ID: 1214647

For an underlying asset with no holding costs or benefits, the no-arbitrage forward price equals:

A) the spot price.
B) the future value of the spot price.

C) zero at initiation of a forward contract.

Explanation

With no costs or benefits to holding the underlying asset, the no-arbitrage forward price is the future value of the spot price, compounded at the risk-free rate over the term of a forward contract.

For Further Reference:

(Study Session 16, Module 49.1, LOS 49.d)

CFA® Program Curriculum, Volume 5, page 469

Question #103 of 120

Question ID: 1214650

Which of the following portfolios has the same future cash flows as a protective put?

A) **Long call option, long risk-free bond.**
B) Short call option, long risk-free bond.
C) Long call option, long risk-free bond, short the underlying asset.

Explanation

The put-call parity relationship shows that a protective put (long put, long underlying asset) has the same future payoff as a fiduciary call (long call, long risk-free bond).

For Further Reference:

(Study Session 16, Module 49.3, LOS 49.l)

CFA® Program Curriculum, Volume 5, page 491

Question #104 of 120

Question ID: 1214651

The right to exercise an option prior to expiration is *most likely* to have value for:

A) a deep in-the-money call option on an asset with no cash flows.
B) an at-the-money put option on an asset with positive cash flows.
C) **an at-the-money call option on an asset with positive cash flows.**

Explanation

It may be valuable to exercise a call option before expiration when the underlying asset makes cash distributions (pays dividend or interest payments). The cash distribution will decrease the price of the underlying asset, decreasing the call value on the date of an interest payment or the ex-dividend date for a dividend payment.

For Further Reference:

(Study Session 16, Module 49.4, LOS 49.o)

CFA® Program Curriculum, Volume 5, page 500

Question #105 of 120

Question ID: 1214648

Which of the following is *least likely* a pure arbitrage strategy?

A) Selling a forward contract and buying the underlying asset.
B) **Borrowing at 1-year LIBOR and buying a 5-year Treasury bond.**
C) Borrowing at 30-day LIBOR, lending at 90-day LIBOR, and buying a 30-day FRA on 60-day LIBOR.

Explanation

Borrowing at a short rate to buy a longer-dated bond exposes the investor to risk from changes in the borrowing rate.

With favorable pricing, selling a forward and buying and storing the underlying asset could generate a riskless profit when the asset is delivered for the forward price.

Borrowing at 30-day LIBOR lending at 90-day LIBOR creates the equivalent of a short position in a 30-day FRA on 60-day LIBOR. If the rate on the long 30-day FRA is high enough, the portfolio will generate a riskless profit.

For Further Reference:

(Study Session 16, Modules 48.2, 49.1, 49.2, LOS 48.f, 49.a, 49.f)

CFA® Program Curriculum, Volume 5, page 430

CFA® Program Curriculum, Volume 5, page 456

CFA® Program Curriculum, Volume 5, page 473

Question #106 of 120

Question ID: 1214646

An investor buys an October 80 put option on a stock for a premium of $5. On the expiration date the stock price is 78 per share. The investor's gain or loss on the option is:

A) a loss of $2.
B) **a loss of $3.**
C) a gain of $2.

Explanation

The intrinsic value of a put is the greater of zero or the difference between the exercise price and the stock price if stock price. This option is in the money by 80 – 78 = 2. The gain or loss is equal to the intrinsic value minus the premium paid: 2 – 5 = –3.

For Further Reference:

(Study Session 16, Module 48.2, LOS 48.d)

CFA® Program Curriculum, Volume 5, page 406

Question #107 of 120

Question ID: 1214652

Questions 107 through 113 relate to Alternative Investments. (10.5 minutes)

For the valuation of real estate investment trusts (REITs), the asset-based approach estimates value by:

A) subtracting recurring capital expenditures from funds from operations.

B) adding depreciation, subtracting gains from property sales, and adding losses on property sales.

C) **subtracting total liabilities from the total value of the real estate assets and dividing by the number of shares outstanding.**

Explanation

The asset-based approach provides an estimate of the net asset value of REITs by subtracting total liabilities from the total value of the real estate assets and dividing by the number of shares outstanding. The income-based approach uses either funds from operations (FFO) or adjusted funds from operations (AFFO) to compute cash flows. FFO is calculated from net income with depreciation added back and with gains from property sales subtracted and losses on property sales added. AFFO is FFO with recurring capital expenditures subtracted.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 10

Question #108 of 120

Question ID: 1214654

On the financial statements of a hedge fund, asset values for positions the fund holds are *most appropriately* calculated using the bid price for:

A) all positions.

B) short positions, and the ask price for long positions.

C) **long positions, and the ask price for short positions.**

Explanation

For financial reporting purposes the most conservative way to value positions is to use the bid for *long* positions and the ask for *short* positions, because these best reflect the prices at which positions may be closed.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.e)

CFA® Program Curriculum, Volume 6, page 10

Question #109 of 120

Question ID: 1214653

The period of time during which a private equity fund will select investments and direct committed capital to them is *best* described as a:

A) notice period.

B) lockup period.

C) **drawdown period.**

Explanation

The drawdown period for a private equity fund is the span of time over which the fund will draw down its committed capital and use it to invest in portfolio companies. Lockup periods and notice periods are related to hedge fund redemptions.

For Further Reference:

(Study Session 17, Module 50.1, LOS 50.d)

CFA® Program Curriculum, Volume 6, page 20

Question #110 of 120

Question ID: 1214655

Arkex Funds is a hedge fund with a value of $100 million at the beginning of the year. Arkex Funds charges a 2.0% management fee based on assets under management at the beginning of the year and a 20.0% incentive fee with a 5.0% hard hurdle rate. Incentive fees are calculated net of management fees. The value of the fund at the end of the year before fees is $110 million. The net return to investors is *closest* to:

A) 6.8%.
B) **7.4%.**
C) 8.0%.

Explanation

Management fee = $100 million × 2.0% = $2.0 million

Incentive fee = [$110 million − $100 million − $2 million − ($100 million × 5.0%)] × 20% = $0.6 million

Total fee = $2.0 million + $0.6 million = $2.6 million

End-of-year value after fees = $110.0 − $2.60 = $107.4 million

Net return = ($107.4 million / $100.0 million) − 1 = 7.4%.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.d)

CFA® Program Curriculum, Volume 6, page 20

Question #111 of 120

Question ID: 1214656

A long-only commodity index investment is *most likely* to:

A) perform poorly in inflationary periods.
B) have a high correlation with equity returns.
C) **be implemented by using derivatives.**

Explanation

Investment in commodities is most frequently implemented using derivatives. Among the important motivations for investing in commodities are that they can serve as an inflation hedge and that their returns typically have a low correlation with equity and bond returns.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.b)

CFA® Program Curriculum, Volume 6, page 10

Question #112 of 120

Question ID: 1214657

Measures of downside risk such as the Sortino ratio are *most likely* to be more appropriate than standard deviation for measuring risk of a:

- **A) macro strategy fund.**
- **B)** real estate investment trust.
- **C)** commodity exchange-traded fund.

Explanation

Downside risk measures may be more appropriate than standard deviation for estimating risk of hedge funds, which often have leptokurtic and negatively skewed returns distributions. Standard deviation is typically an appropriate risk measure for publicly traded securities such as REITs and ETFs.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.f)

CFA® Program Curriculum, Volume 6, page 57

Question #113 of 120

Question ID: 1214658

Which of the following is *least likely* an important element of risk management for alternative investment funds?

- **A)** Setting leverage limits.
- **B)** Obtaining independent valuations of fund positions.
- **C) Designating a risk management officer who also works as part of the investment team.**

Explanation

The risk officer should be separate from those who manage the investment process. Setting leverage limits and getting third party (not in-house) valuations of fund investments are both important elements of risk management for alternative investment funds.

For Further Reference:

(Study Session 17, Module 50.2, LOS 50.f)

CFA® Program Curriculum, Volume 6, page 57

Question #114 of 120

Question ID: 1214665

Questions 114 through 120 relate to Portfolio Management. (10.5 minutes)

The sensitivity of a derivative's value to the price of the underlying asset is measured by the derivative's:

- **A)** beta.

B) delta.

C) gamma.

Explanation

Delta measures the sensitivity of a derivative's value to the price of its underlying asset. Gamma measures the sensitivity of delta to the price of the underlying asset. Beta measures the market risk of a security.

For Further Reference:

(Study Session 19, Module 55.1, LOS 55.g)

CFA® Program Curriculum, Volume 6, page 315

Question #115 of 120

Question ID: 1214661

The standard deviation of returns for a portfolio of risky assets is:

A) less than the standard deviation of the least risky asset.

B) greater than the standard deviation of the least risky asset.

C) less than or equal to the standard deviation of the most risky asset.

Explanation

The standard deviation of return of a portfolio of many risky assets can be less than or greater than the standard deviation of the least risky asset, depending on the correlations of returns among the assets and their weights in the portfolio. However, the portfolio standard deviation must be less than or equal to the standard deviation of the most risky asset.

For Further Reference:

(Study Session 18, Module 52.3, LOS 52.f)

CFA® Program Curriculum, Volume 6, page 150

Question #116 of 120

Question ID: 1214659

An equally weighted portfolio of five securities has a standard deviation of returns of 10%. The average standard deviation of returns of the five securities is 15%. If another security with a standard deviation of returns of 15% is added to the portfolio, and the weights are adjusted to restore equal weighting, the portfolio's diversification ratio is *most likely* to:

A) increase.

B) decrease.

C) remain unchanged.

Explanation

The diversification ratio is the standard deviation of an equally weighted portfolio's returns divided by the average standard deviation of returns of the securities in the portfolio. Adding a security with a standard deviation equal to the average will decrease the portfolio standard deviation as long as the added security's returns are not perfectly positively correlated with the portfolio's returns. The result is a lower diversification ratio, which indicates a greater benefit from diversification.

For Further Reference:

(Study Session 18, Module 51.1, LOS 51.a)

CFA® Program Curriculum, Volume 6, page 80

Question #117 of 120

Question ID: 1214663

Which of the following statements about the security market line (SML) is *least accurate*?

A) The independent variable in the SML equation is the standard deviation of the market portfolio.

B) The SML measures risk using the standardized covariance of the stock with the market.

C) Securities plotting above the SML are undervalued.

Explanation

The SML uses either the covariance between assets and the market or beta as the measure of risk. Beta is the covariance of a stock with the market divided by the variance of the market. Securities that plot above the SML are undervalued and securities that plot below the SML are overvalued.

For Further Reference:

(Study Session 18, Module 53.2, LOS 53.f, 53.h)

CFA® Program Curriculum, Volume 6, page 213

CFA® Program Curriculum, Volume 6, page 217

Question #118 of 120

Question ID: 1214664

The problem of investment managers taking offsetting active positions is *most likely* addressed by employing:

A) risk budgeting.

B) tactical asset allocation.

C) a core-satellite approach.

Explanation

With a core-satellite approach, a majority of the assets are invested passively (in the "core" portfolio) with a smaller proportion in actively managed "satellite" portfolios. This reduces the likelihood of excessive trading and offsetting active positions.

For Further Reference:

(Study Session 19, Module 54.1, LOS 54.g)

CFA® Program Curriculum, Volume 6, page 265

Question #119 of 120

Question ID: 1214662

Bearing unsystematic risk should provide no additional expected return:

A) under any circumstances.

B) **if diversification is cost-free.**

C) in a strong-form efficient market.

Explanation

The theory that bearing unsystematic risk will provide no additional expected return assumes that unsystematic risk can be diversified away at no cost.

For Further Reference:

(Study Session 12, Module 38.1, LOS 38.d and Study Session 18, Module 53.1, LOS 53.c)

CFA® Program Curriculum, Volume 4, page 326

CFA® Program Curriculum, Volume 6, page 204

Question #120 of 120

Question ID: 1214660

An investor buys a non-dividend paying stock for $100 at the beginning of the year with 50% initial margin. At the end of the year, the stock price is $95. Deflation of 2% occurred during the year. Which of the following return measures for this investment will be greatest?

A) **Real return.**

B) Nominal return.

C) Leveraged return.

Explanation

No calculations are needed. The real return is greater than the nominal return because the inflation rate is negative. The leveraged return is more negative than the nominal return because the investment lost value and leverage magnifies the loss.

For Further Reference:

(Study Session 18, Module 52.1, LOS 52.a)

CFA® Program Curriculum, Volume 6, page 116